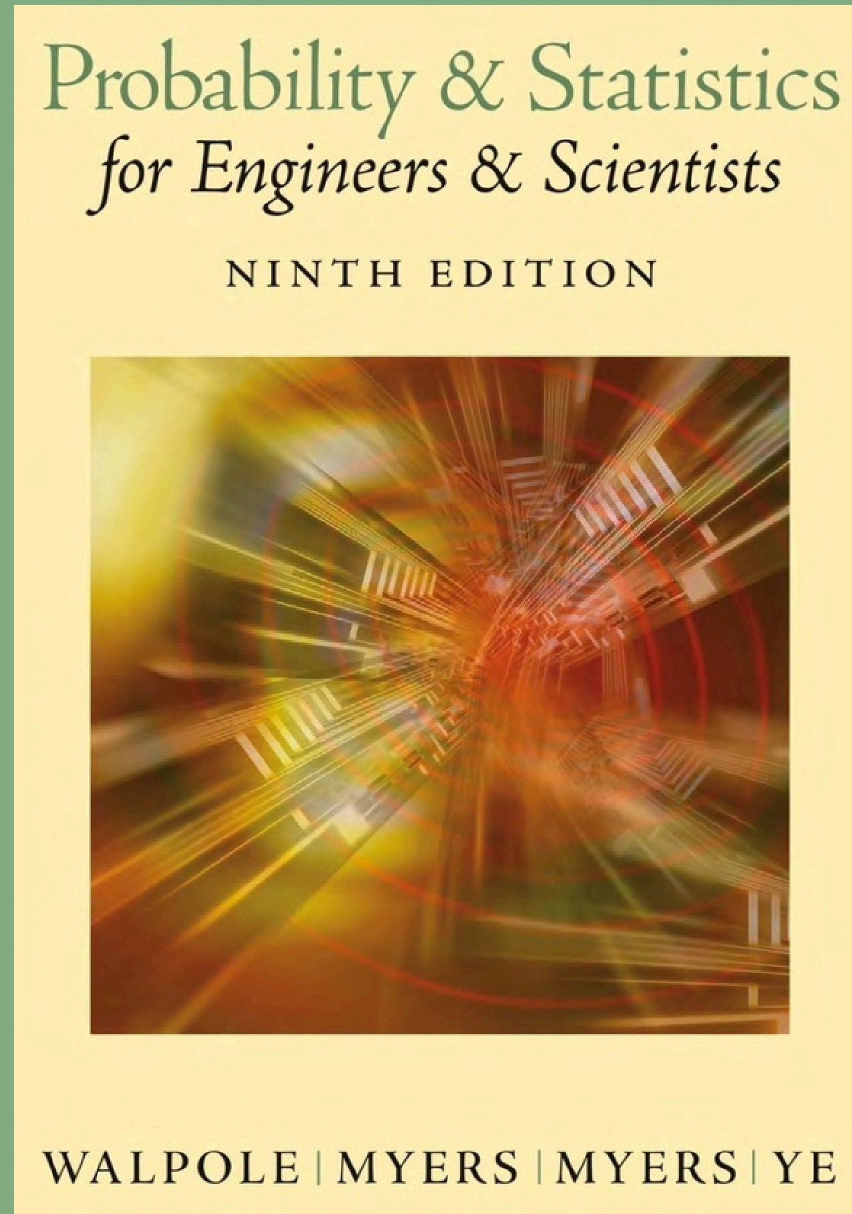
Chapter 3

Random Variables



Section 3.1

Concept of a Random Variable

Definition 3.1

A **random variable** is a function that associates a real number with each element in the sample space.

Example: Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. The random variable Y is the number of red balls.

Definition 3.2

If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a **discrete sample space**.

Definition 3.3

If a sample space contains an infinite number of possibilities equal to the number of points on a line segment, it is called a **continuous sample space**.

Section 3.2

DiscreteProbability Distribution

Definition 3.4

The set of ordered pairs $(x, f(x))$ is a **probability function**, **probability mass function**, or **probability distribution** of the discrete random variable X if, for each possible outcome x,

1. $f(x) \geq 0$,
2. $\sum_{x} f(x) = 1$,
3. $P(X = x) = f(x)$.

Definition 3.5

The **cumulative distribution function** $F(x)$ of a discrete random variable X with probability distribution $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \quad \text{for } -\infty < x < \infty.$$

Example: If a car agency sells 50% of its inventory of a certain foreign car equipped with side airbags, find a formula for the probability distribution of the number of cars with side airbags among the next 4 cars sold by the agency.

Figure 3.1 Probability mass function plot

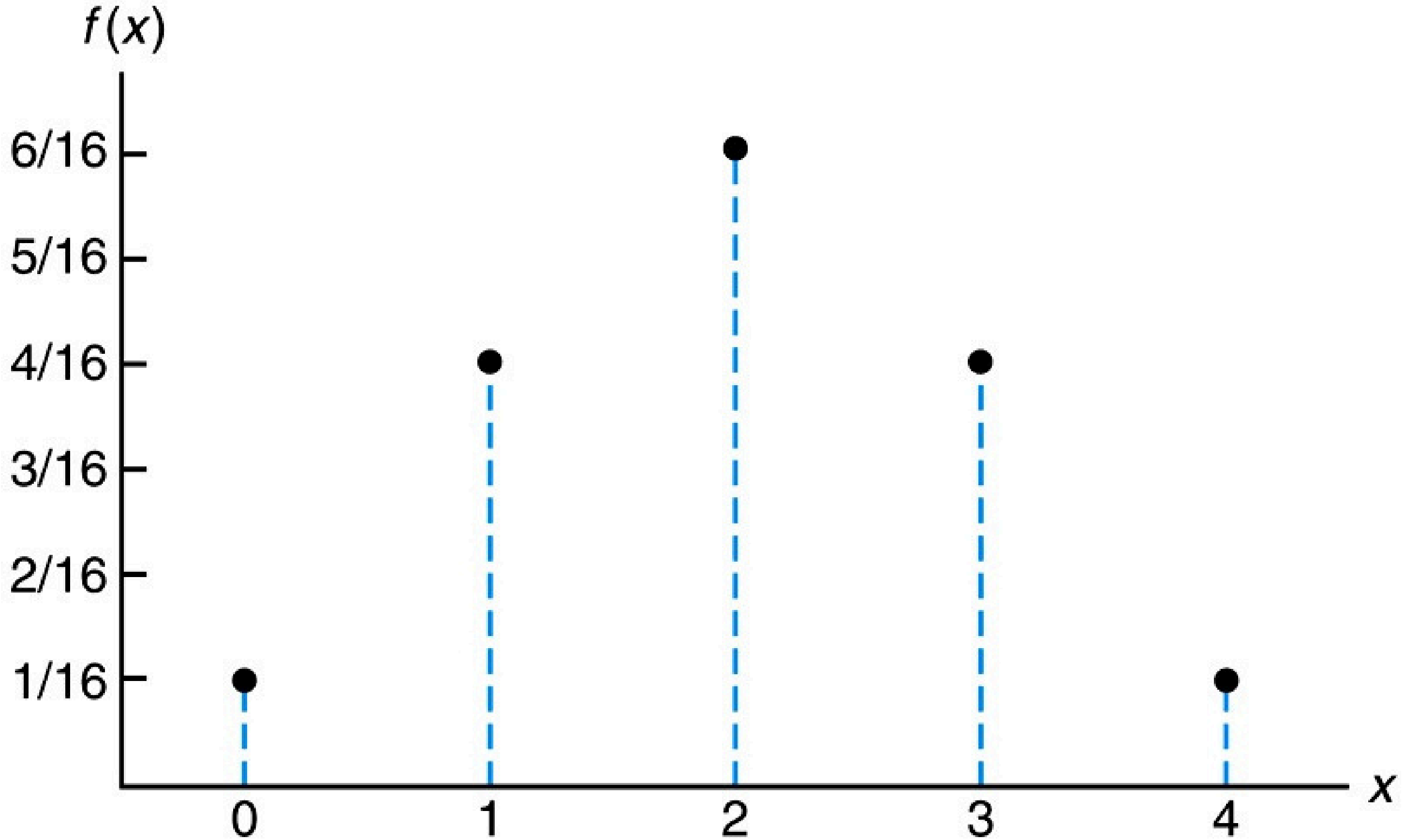


Figure 3.2 Probability histogram

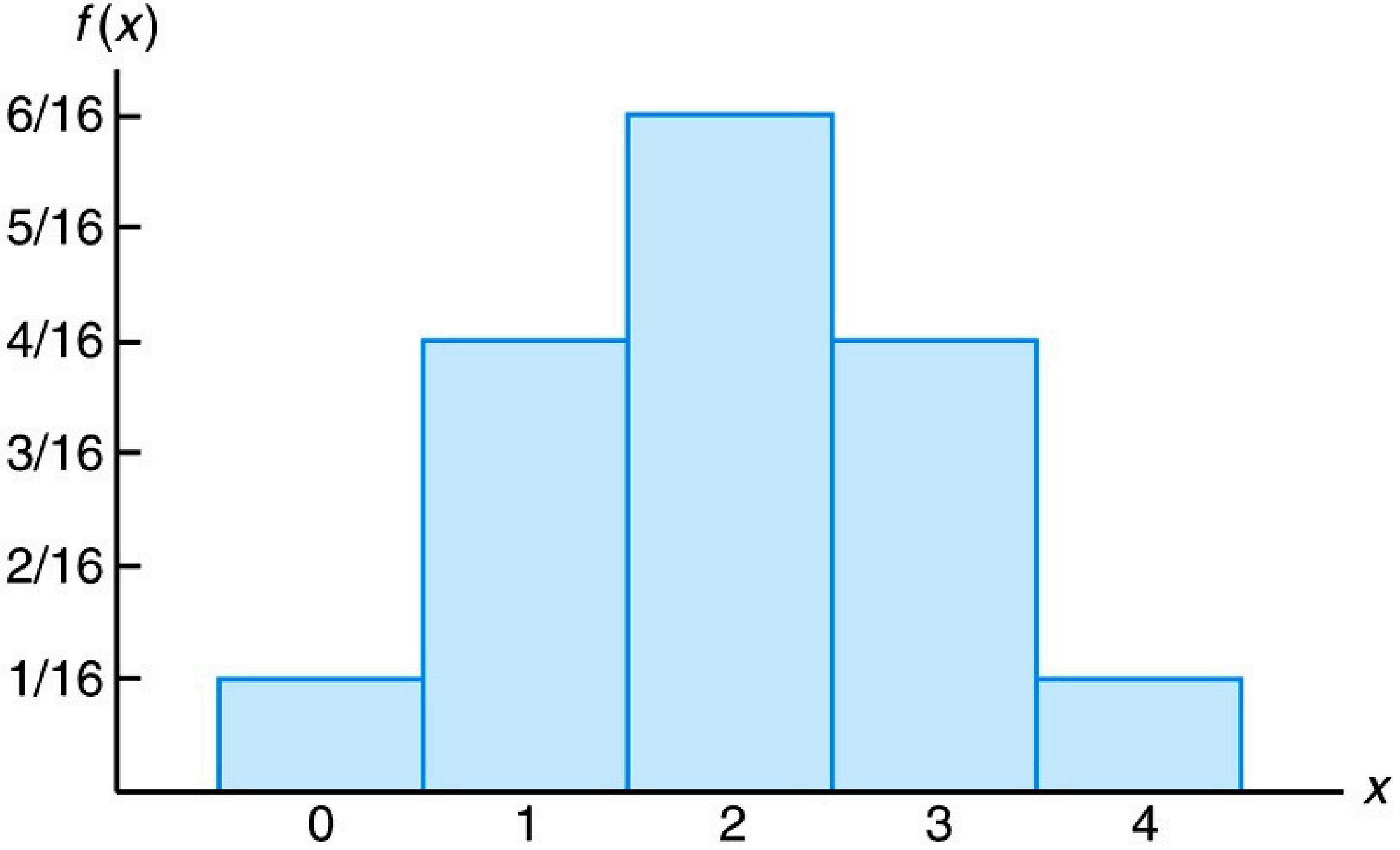
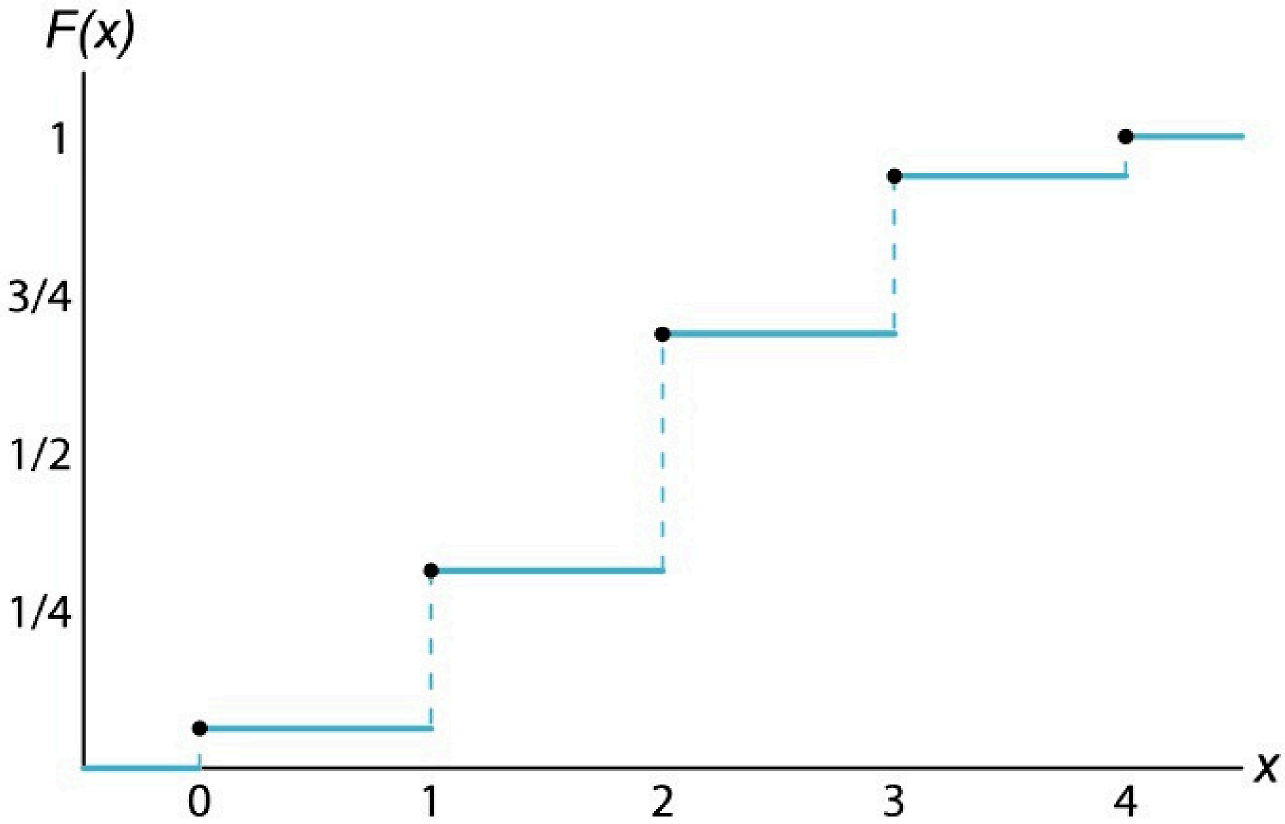


Figure 3.3 Discrete cumulative distributionfunction



Section 3.3

ContinuousProbability Distributions

Figure 3.4 Typical density functions

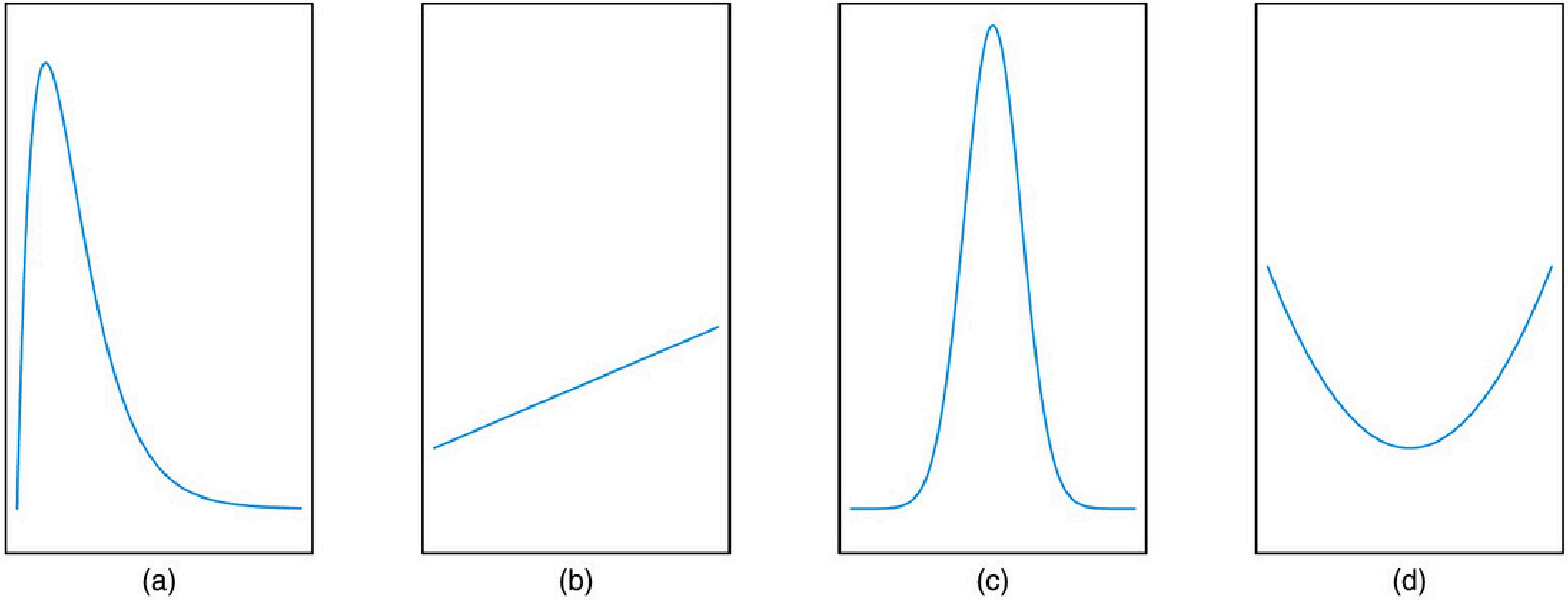
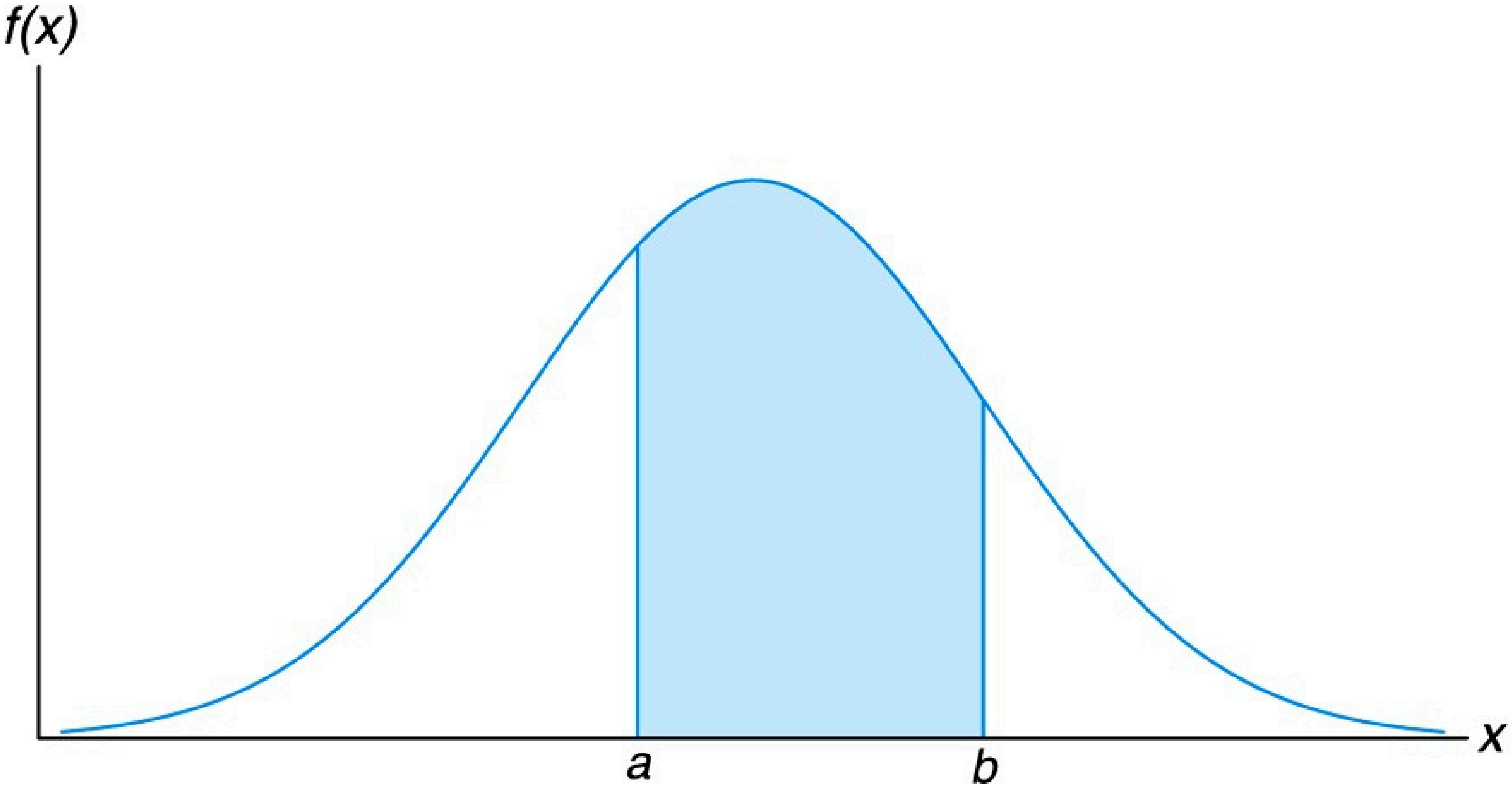


Figure 3.5 $P(a < X < b)$



Definition 3.6

The function $f(x)$ is a **probability density function** (pdf) for the continuous random variable X, defined over the set of real numbers, if

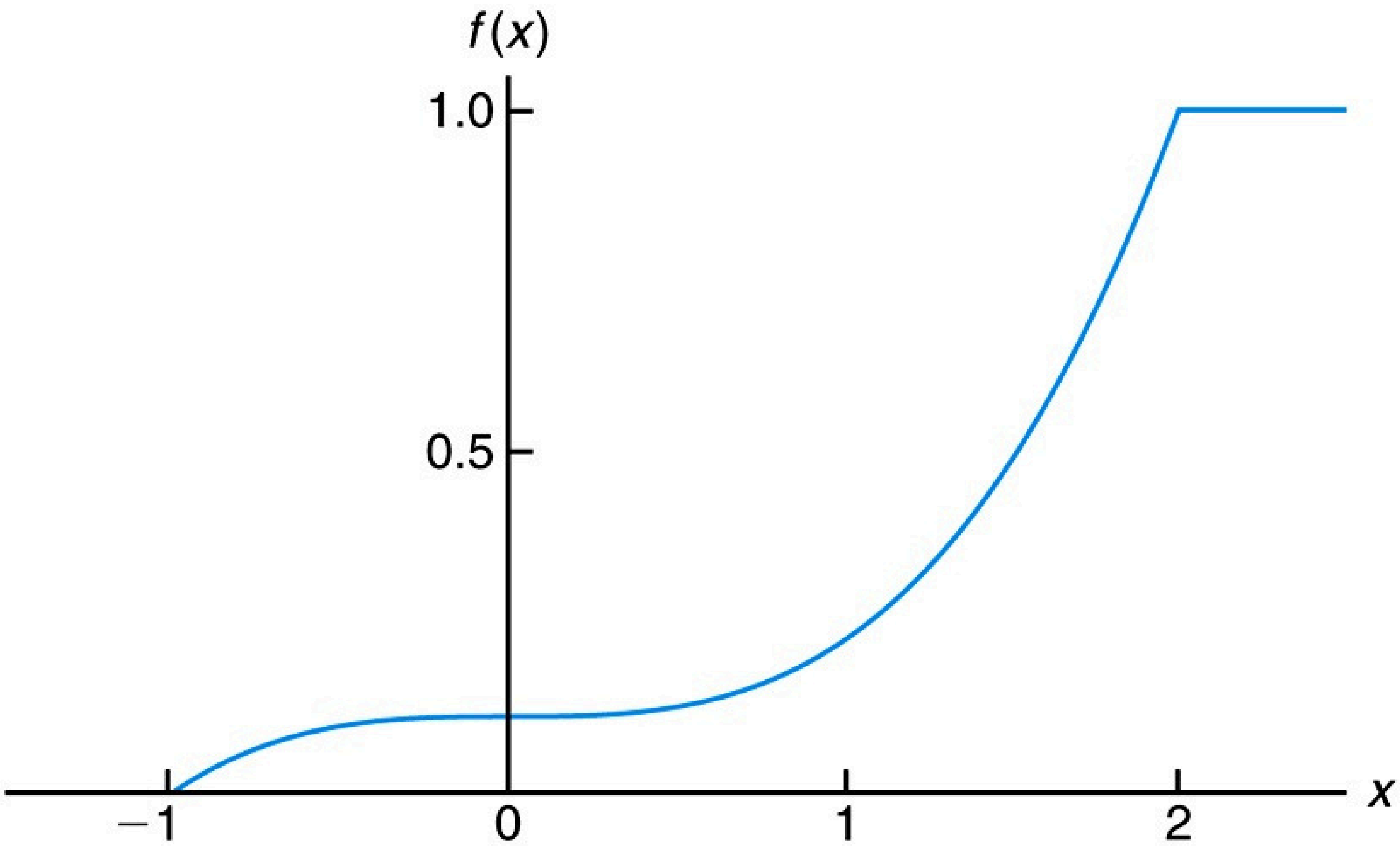
1. $f(x) \geq 0$, for all $x \in R$.
2. $\int_{-\infty}^{\infty} f(x)\, dx = 1$.
3. $P(a < X < b) = \int_a^b f(x)\, dx$.

Definition 3.7

The **cumulative distribution function** $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt, \quad \text{for } -\infty < x < \infty.$$

Figure 3.6 Continuous cumulative distributionfunction



EXERCISE

3.40 A fast-food restaurant operates both a drive-through facility and a walk-in facility. On a randomly selected day, let X and Y, respectively, be the proportions of the time that the drive-through and walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x,y) = \begin{cases} \frac{2}{3}(x+2y), & 0 \le x \le 1,\ 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find the marginal density of X.

3.43 Let X denote the reaction time, in seconds, to a certain stimulus and Y denote the temperature (°F) at which a certain reaction starts to take place. Suppose that two random variables X and Y have the joint density

$$f(x,y) = \begin{cases} 4xy, & 0 < x < 1,\ 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find

(a) $P(0 \le X \le \frac{1}{2}$ and $\frac{1}{4} \le Y \le \frac{1}{2})$;

(b) $P(X < Y)$.

Find

(a) $P(0 \le X \le \frac{1}{2}$ and $\frac{1}{4} \le Y \le \frac{1}{2})$;

(b) $P(X < Y)$.

Solution
3.40

$$f(x,y) = \begin{cases} \frac{2}{3}(x+2y), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$g(x) = \int_0^1 f(x,y)\,dy = \int_0^1 \frac{2}{3}(x+2y)\,dy$$

$$g(x) = \frac{2}{3}\left[xy + y^2\right]_{y=0}^{y=1}$$

$$g(x) = \frac{2}{3}(x(1) + (1)^2 - 0) = \frac{2}{3}(x+1), \quad 0 \leq x \leq 1$$

Solution
3.43

$$h(y) = \int_0^1 f(x,y)\,dx = \int_0^1 \frac{2}{3}(x+2y)\,dx$$

$$h(y) = \frac{2}{3}\left[\frac{x^2}{2} + 2yx\right]_{x=0}^{x=1}$$

b) $$h(y) = \frac{2}{3}\left(\frac{1}{2} + 2y\right) = \frac{1}{3} + \frac{4}{3}y = \frac{1}{3}(1+4y), \quad 0 \le y \le 1$$

EXERCISE

3.66 Consider the random variables X and Y with joint density function

$$f(x, y) = \begin{cases} x + y, & 0 \leq x, y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find the marginal distributions of X and Y.

(b) Find $P(X > 0.5, Y > 0.5)$.

3.62 An insurance company offers its policyholders a number of different premium payment options. For a randomly selected policyholder, let X be the number of months between successive payments. The cumulative distribution function of X is

$$F(x) = \begin{cases} 0, & \text{if } x < 1, \\ 0.4, & \text{if } 1 \leq x < 3, \\ 0.6, & \text{if } 3 \leq x < 5, \\ 0.8, & \text{if } 5 \leq x < 7, \\ 1.0, & \text{if } x \geq 7. \end{cases}$$

Solution
3.66

Joint Density: $f(x,y) = x + y$ for $0 \le x, y \le 1$.

Find the marginal distributions of X and Y

For X: $g(x) = \int_0^1 (x+y)\,dy = [xy + \frac{y^2}{2}]_0^1 = x + \frac{1}{2}$

For Y: $h(y) = \int_0^1 (x+y)\,dx = [\frac{x^2}{2} + xy]_0^1 = y + \frac{1}{2}$

(b) Find $P(X > 0.5, Y > 0.5)$

$$P = \int_{0.5}^{1}\int_{0.5}^{1}(x+y)\,dx\,dy = \int_{0.5}^{1}\left[\frac{x^2}{2} + xy\right]_{0.5}^{1} dy = \int_{0.5}^{1}\left((\frac{1}{2} + y) - (\frac{1}{8} + 0.5y)\right) dy$$

$$P = \int_{0.5}^{1}(0.5y + 0.375)\,dy = [0.25y^2 + 0.375y]_{0.5}^{1} = (0.25 + 0.375) - (0.0625 + 0.187$$

Solution

3.66

$$F(x) = \begin{cases} 0, & x < 1 \\ 0.4, & 1 \le x < 3 \\ 0.6, & 3 \le x < 5 \\ 0.8, & 5 \le x < 7 \\ 1.0, & x \ge 7 \end{cases}$$

Verification (Sum of Probabilities):

$$\sum P(X = x_i) = P(X = 1) + P(X = 3) + P(X = 5) + P(X = 7)$$

$$\sum P(X = x_i) = 0.4 + 0.2 + 0.2 + 0.2 = \mathbf{1.0}$$

Probability Mass Function (PMF):

$$P(X = x) = F(x) - F(x^-)$$

$$P(X = 1) = F(1) - F(1^-) = 0.4 - 0 = \mathbf{0.4}$$

$$P(X = 3) = F(3) - F(3^-) = 0.6 - 0.4 = \mathbf{0.2}$$

$$P(X = 5) = F(5) - F(5^-) = 0.8 - 0.6 = \mathbf{0.2}$$

$$P(X = 7) = F(7) - F(7^-) = 1.0 - 0.8 = \mathbf{0.2}$$